**UNIVERSITI MALAYSIA SARAWAK**

Faculty of Computer Science and Information Technology

TMA3084 Software Engineering Laboratory | Semester 1, 2026/2027

# PROJECT PROPOSAL

*(Product Requirements Document)*

## KerepekHub: Ordering & Customer Feedback Application (KerepekHub)

| | |
|---|---|
| **Group Number / Name** | Group 16 - LadyLogics |
| **Lab Group / Demonstrator** | Group 2 –Mr. Raynold anak Kabai |
| **Client / Organisation** | Nur Asyiqin, Founder, Serooja |
| **Lecturer** | Mdm. Nurfauza Jali |
| **GitHub Repository** | https://github.com/damiaqistina/KerepekHub |
| **Version / Date** | v1.0 (original) – dd/10/2026 |

### Team Members

| No. | Name | Matric No. | TSPi Role | Signature |
|---|---|---|---|---|
| *1* | SNG NIC CO | 106420 | Development Manager | |
| *2* | AINUR HANUM ABDULLAH TASIP | 103158 | Quality/ Process Manager | |
| *3* | SITI AISYAH BINTI SABEKI | 106351 | Planning Manager | |
| *4* | DAMIA QISTINA BINTI FAIZAL | 102504 | Project Manager | |
| *5* | SABRINA NATASHA BINTI SHUDARSAN | 106199 | Lead Mobile Architect | |

**Declaration**

By signing above, we certify that this proposal is our own work, that all external sources, libraries, APIs and AI-assisted content are acknowledged, and that we have not engaged in plagiarism or unauthorised collusion (Peraturan Akademik UNIMAS, Etika Akademik). AI tool usage is recorded in LOGAI.

## How to Use This Template (delete this page before submission)

*‣ Grey italic text (‣) and blue 'PRD section' bars are guidance. Replace them with your own content, then delete them.*

*‣ Target length: 10–15 pages excluding cover and appendices. Be concise: tables beat long paragraphs.*

*‣ Formatting: Times New Roman 12 pt, 1.5 line spacing, left aligned. Heading 1 = 14 pt bold, Heading 2 = 12 pt bold. Right-click the Table of Contents → Update Field before submitting.*

*‣ Every figure and table needs a numbered caption and must be referred to in the text. Add alt text to images.*

*‣ Keep IDs consistent: the FR-xx IDs you define here are reused in the SRS, GitHub Issues, test cases (LOGTEST) and LOGAI.*

### Proposal Checklist → PRD Mapping

| Proposal checklist item | PRD equivalent | Key question it answers |
|---|---|---|
| Abstract | Executive Summary / TL;DR | What are we building, for whom, and why? |
| Introduction | Background & Context | Who is the client and who is the team? |
| Problem Statement | Problem & Opportunity | What pain exists today and what is the evidence? |
| Project Objectives | Goals & Success Metrics | How will we know the project succeeded? |
| Project Perspective | Users/Personas & System Context | Who uses it and how does it fit with other systems? |
| Project Scope | Scope, Features & Requirements | What is in, what is out, and what is the AI feature? |
| Task Allocation + Team Contribution | Team / Stakeholders (RACI) | Who owns what? |
| Team Goals | Team Working Agreement | How will the team work and measure itself? |
| Expected Result | Release Criteria / Deliverables | What will be handed over at the end? |
| Project Planning | Milestones, Timeline, Risks | When, in what order, and what could go wrong? |

**Table A: Mapping of proposal checklist to PRD sections**

## Table of Contents


Abstract....6
1. Introduction....6
    1.1 Client Background....6
    1.2 Team Background....7
2. Problem Statement....7
3. Project Objectives & Success Metrics....8
4. Project Perspective....9
    4.1 Target Users....9
    4.2 System Context and Architecture....10
5. Project Scope....11
    5.1 In Scope / Out of Scope....11
    5.2 Features and User Stories....12
    5.3 AI Feature....14
    5.4 Non-Functional Requirements and Constraints....15
    5.5 System Environment....16
6. Task Allocation & Team Contribution....17
7. Team Goals....18
8. Expected Results....19
    8.1 Working System....19
    8.2 Documentation and Artefacts....20
9. Project Planning....20
    9.1 Methodology....20
    9.2 Submission Schedule....21
    9.3 Development Milestones (GitHub)....22
    9.4 Risks and Assumptions....22
10. References....23
Appendix A: Client Meeting Evidence....24
Appendix B: Revision History....24

## Abstract

KerepekHub is a cross-platform flutter mobile application designed to digitize and streamline Serooja, an independent micro-business managed by its founder, Nur Asqyiqin. Currently, Serooja relies on traditional manual ordering through WhatsApp and stock tracking via paper notebooks, resulting in longer stock management and stock oversights. Kerepekhub mitigates these issues by offering dual role capabilities. Customers can browse, place orders, select self-collection or delivery, and track real time order progression. Simultaneously, the business owner can manage the incoming orders, update inventory quantities, verify payment status, and view sales performance analytics. Powered by Google Gemini AI, the system generates automated sales trend insights, highlights low-stock items, and provides stock replenishment priorities on the seller dashboard. Built using Flutter, Supabase, and GitHub Actions CI/CD, KerepekHub establishes a scalable, digital foundation to support Serooja's sustainable business growth.

## 1. Introduction

KerepekHub: Ordering & Customer Feedback Application is a software engineering project developed by team LadyLogics to digitize and optimize daily business operations for Serooja, a local independent business. Managed by its founder, Nur Asyiqin, Serooja specializes in several distinct snack varieties. The primary goal of this project is to replace Seroja's current reliance on manual stock tracking and unstructured WhatsApp messages with a streamlined, dual-role mobile application tailored for both customers and the business owner's needs.

A mobile application built on the Flutter framework is the ideal solution for Serooja due to its accessibility, real-time capabilities, and cross-platform flexibility. For Serooja's customer base, which consists primarily of tech-literate adults, an application that provides the convenience of browsing products and its availability sets Serooja apart from its similar-scale competitors. For a business owner, a dedicated mobile application centralizes orders, enables instantaneous inventory updates, and provides sales analytics. This strategy eliminates missed orders and repetitive customer enquiries, particularly about the availability of the products. This positions Serooja for sustainable business growth.

### 1.1 Client Background

Serooja is an independent local snack (kerepek) business founded by Nur Asyiqin. Having operated her kerepek business since age 16, the owner established Serooja as the go-to snack for her loyal customers, offering several variations of kerepek such as Crunchy Spicy Cheese, Crunchy Nestum Milk, Seaweed Original, Seaweed Spicy Cheese, and a limited-edition product called Popia Simpul, during peak season such as Eid al-Fitr. Products are distributed in two distinct sizes, large bottle containers and smaller packet portions. The primary target customer

base consists of students, working adults, and families purchasing snacks for their personal consumption or as gifts.

### 1.2 Team Background

LadyLogics comprises five third-year Software Engineering students with coursework experience in cross-platform mobile development, databases, and cloud Generative AI integration. The team stands by the motto "Connecting Code to Local Communities."

## 2. Problem Statement

Serooja currently manages its business orders manually through WhatsApp messages and notebook. As the business grows and receives more orders, the owner, Nur Asyiqin faces difficulties keeping track of pending and completed orders, updating stock quantities, and ensuring that customers receive the correct products. Based on the client interview, three main problems have been identified.

| ID | Current process / problem | Impact on client | Evidence (source) |
|---|---|---|---|
| P1 | Orders are managed manually through Whatsapp messages and notes, making it difficult to track multiple orders at once. | Orders may be missed, forgotten, or recorded with incorrect quantities. The owner also spends more time checking and confirming orders. | The client uses WhatsApp and a notebook to track orders, which may lead to missed orders when handling multiple orders. *(Client interview, 6/10/2026)* |
| P2 | Product stock is checked and updated manually using a notebook. | The owner may have difficulty maintaining accurate stock records, and customers may order products that are already out of stock | The client records stock manually and confirmed that customers have sometimes ordered products that were already out of stock. *(Client interview, 6/10/2026)* |
| P3 | Customers need to contact the owner to ask about product availability, prices, | Customers may experience inconvenience, while the owner spends additional time answering | Customers ask about product availability, prices, flavours, and order status, requiring the client to |

| ID | Current process / problem | Impact on client | Evidence (source) |
|---|---|---|---|
| | flavours, and order status. | repeated enquiries and checking order updates. | spend additional time responding to enquiries. *(Client interview, 6/10/2026)* |

**Table 1: Summary of client problems**

To address these problems, the team proposes developing KerepekHub, a Flutter-based mobile application for Serooja that allows customers to browse products and place orders while enabling the owner to manage orders, update stock, and monitor order statuses through a more organised system. The application aims to reduce manual errors, improve ordering convenience, and support the future growth of Serooja.

## 3. Project Objectives & Success Metrics

| No. | Objective | Solves | Success metric (how we measure) | Priority |
|---|---|---|---|---|
| *O1* | To study and analyse the client's current business processes such as order and stock management, also monitor sales performance. | All | SRS signed off by client by Milestone 2. | High |
| *O2* | To design and develop a Flutter app that allows customers to see product availability, place orders and monitor status. Allows the client as the seller to manage and record orders and sales, update product stock and order status. | P1, P2 | All Must-have FRs pass acceptance tests. | High |
| *O3* | To integrate an AI feature that analyzes sales and stock data for the seller dashboard displaying sales trends and notify about product restocks. | P2 | ≥80% of predefined test scenarios producing accurate stock and sales insights. | High |
| *O4* | To test and evaluate application functionality and receive client | All | ≥50 completed test cases, ≥60% achieved test | Medium |

| No. | Objective | Solves | Success metric (how we measure) | Priority |
|---|---|---|---|---|
| *O1* | To study and analyse the client's current business processes such as order and stock management, also monitor sales performance. | All | SRS signed off by client by Milestone 2. | High |
| *O2* | To design and develop a Flutter app that allows customers to see product availability, place orders and monitor status. Allows the client as the seller to manage and record orders and sales, update product stock and order status. | P1, P2 | All Must-have FRs pass acceptance tests. | High |
| *O3* | To integrate an AI feature that analyzes sales and stock data for the seller dashboard displaying sales trends and notify about product restocks. | P2 | ≥80% of predefined test scenarios producing accurate stock and sales insights. | High |
| | confirmation as business requirements have all been fulfilled. | | coverage, UAT sign-off from client. | |

**Table 2: Project objectives and success metrics**

# 4. Project Perspective

## 4.1 Target Users

| User type | Description | Main needs / goals | Device / tech level |
|---|---|---|---|
| Customer | Individuals who wish to conveniently order items and browse Serooja's products | Browse products, view prices, place orders and track order status | Android 8.0+ phones |
| Admin (Client) | Serooja's owner responsible for | Manage orders, update order status, add, update and delete products and | Android 8.0+ phones |

| User type | Description | Main needs / goals | Device / tech level |
|---|---|---|---|
| | managing products and customer orders | monitor customer purchases | |

**Table 3: Target users**

### 4.2 System Context and Architecture

Figure 1: KerepekHub high-level system architecture

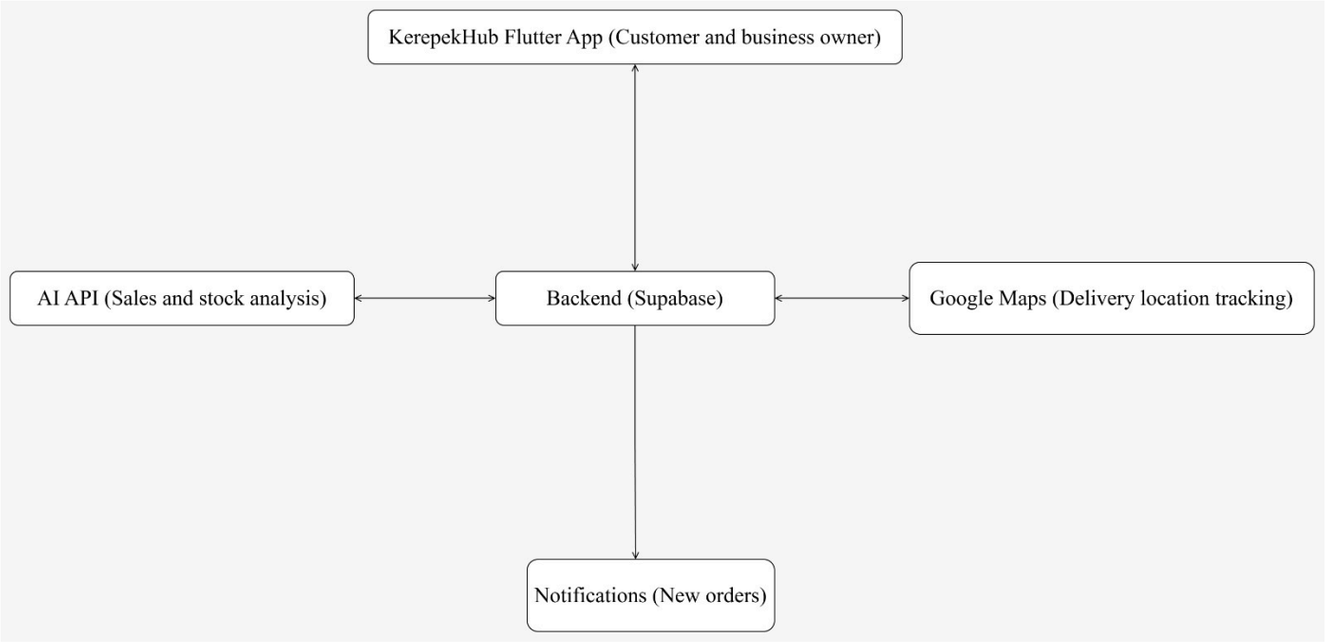



Figure 1: KerepekHub high-level system architecture

KerepekHub is a new mobile ordering system that replaces the business owner's manual process of taking orders and recording sales by hand. The Flutter app is used by both customers and the business owner. In this system, the Supabase backend, which manages authentication, product and stock data, orders, order statuses, payment status records, sales records and customer feedback, becomes the only source of communication between the Flutter app and all other services. The backend connects to an AI API that analyses sales and stock data for the owner's dashboard. It also connects to Google Maps to track delivery locations and a notification service to alert the owner of new orders. Payments are made outside of the app, and the owner confirms them in the app. Keeping all external calls behind the backend helps protect API keys and allows individual services to be replaced without changing the app.

# 5. Project Scope

## 5.1 In Scope / Out of Scope

| In scope (will deliver this semester) | Out of scope (will not deliver) |
|---|---|
| *Customer registration and login* | *Integration with third-party courier services* |
| *Product browsing and viewing product details, flavours, sizes, and prices* | *Multi-vendor marketplace support* |
| *Order placement through the mobile application* | *Integration with bank payment gateway* |
| *Order management for business owner* | *AI voice assistant/chatbot* |
| *Order status tracking (pending, preparing, out for deliver, delivered)* | *Accounting and tax management system* |
| *Delivery location tracking using Google Maps* | *Advanced route optimization* |
| *Product stock management and availability updates* | *iOS specific optimization* |
| *Customer order history viewing* | *Advanced marketing campaign* |
| *Manual payment receipt submission and payment verification* | *Offline application usage* |
| *Order cancellation for unpaid orders after a specific period* | *Automatic payment processing* |
| *Notification for new orders* | *Warehouse inventory management* |
| *Sales record and sales summary viewing* | *Loyalty and reward programmes* |
| *Customer feedback and rating submission* | *Third party logistics management* |
| *AI-based sales and stock analysis dashboard* | *Predictive demand forecasting with external databases* |

**Table 4: Scope boundary**

The scope of KerepekHub is to focuses on some additional features like ordering and business management processes by providing a mobile application for customers and the business owner. Features unrelated to the client's identified problems or exceed the project timeframe are considered outside the scope of this semester's development.

## 5.2 Features and User Stories

| FR-ID | Module | User story (As a … I want … so that …) | Priority |
|---|---|---|---|
| FR-01 | User Management | As a customer, I want to register an account so that I can place and manage my orders securely. | Must |
| FR-02 | User Management | As a customer, I want to log in to my account so that I can access my order history. | Must |

| FR-ID | Module | User story (As a … I want … so that …) | Priority |
|---|---|---|---|
| FR-03 | Product Catalog | As a customer, I want to browse available products, flavours, sizes, and prices so that I can choose products easily. | Must |
| FR-04 | Product Catalog | As a customer, I want to search products so that I can quickly find specific items. | Should |
| FR-05 | Ordering & Cart | As a customer, I want to add products to a cart so that I can order multiple products at once. | Must |
| FR-06 | Ordering & Cart | As a customer, I want to place an order through the application so that I do not need to order manually through WhatsApp. | Must |
| FR-07 | Delivery Management | As a customer, I want to choose either self collection or deliver to the door step so that I can receive my order according to my preference. | Must |
| FR-08 | Order Tracking | As a customer, I want to check my order status so that I know whether my order is preparing/delivering/ready for collection/completed. | Must |
| FR-09 | Order History | As a customer, I want to view my previous orders so that I can reorder products easily. | Should |
| FR-10 | Order Management | As a business owner, I want to view incoming orders so that I can process them efficiently. | Must |
| FR-11 | Order Management | As a business owner, I want to update order status so that customers can track their orders. | Must |
| FR-12 | Stock Management | As a business owner, I want to manage product stock levels so that inventory remains accurate. | Must |
| FR-13 | Product Management | As a business owner, I want to add, edit, or remove products so that product information remains up to date. | Must |
| FR-14 | Payment Management | As a business owner, I want to review uploaded payment receipts and verify payment status so that customer orders can be processed accurately. | Should |

| FR-ID | Module | User story (As a … I want … so that …) | Priority |
|---|---|---|---|
| FR-15 | Customer Management | As a business owner, I want to view customer information and order history so that I can manage customer records efficiently. | Should |
| FR-16 | Sales Reporting & Analytics | As the business owner, I want to view sales summaries and reports so that I can monitor business performance. | Must |
| FR-17 | Notification System | As the business owner, I want to receive notifications when a new order is placed so that I do not miss customer orders. | Should |
| FR-18 | Maps Tracking | As a customer, I want to view the delivery location on a map so that I can collect my order on the right place. | Should |
| FR-19 | Maps Tracking | As a customer, I want to view the seller's real time location on Google Maps so that I can estimate delivery distance and arrival time. | Could |
| FR-20 | Customer Feedback | As a customer, I want to submit feedback and ratings for my completed orders so that I can share my experience with business owner. | Should |
| FR-AI-01 | AI Sales Insight | As a business owner, I want AI-generated sales and stock insights so that I can identify best selling products and required stock replenishment. | Must |

**Table 5: Features and user stories**

*[Insert Figure 2: System modules diagram]*

### 5.3 AI Feature

The proposed of this AI feature helps the business owner make better inventory and sales decision by analysing sales records, order history and product stock data. Instead of displaying raw sales data, the AI generates actionable insights such as identifying best selling products, highlighting low stock items, and suggesting restocking priorities. This feature directly solved P2, where stock levels are currently monitored manually, helping improve inventory management efficiency and business decision making.

| Item | Description |
|---|---|
| AI capability | Sales trend analysis,  inventory recommendation and business insight generation |
| User value | Solves P2 (manual stock management) for the business owner by identifying sales patterns and recommending stock replenishment actions |
| Input → Output | Sales records, order history, stock levels → Best selling products, sales trends, low-stock alerts, restock recommendations |
| AI service / API | Google Gemini API |
| Constraints | ➔ Cost Constraint: Subject to Gemini free-tier API limits<br>➔ Security Constraint: Gemini API key will be stored securely using environment variables and excluded from GitHub repository<br>➔ Privacy Constraint: No sensitive banking information will be sent to the AI service<br>➔ Fallback Constraint: Basic sales reports remain available if the AI service is unavailable |

**Table 6: AI feature summary**

### 5.4 Non-Functional Requirements and Constraints

| NFR-ID | Category | Requirement / constraint | Target |
|---|---|---|---|
| NFR-01 | Performance | The application shall display main screens efficiently. | Main screens load within 5 seconds on a mid-range Android device. |
| NFR-02 | Usability | Customers shall be able to browse products and place orders easily without training. | At least 80% of users successfully complete an order during user testing. |
| NFR-03 | Security | User accounts shall be protected through secure authentication. | Supabase Authentication is required before accessing account features. |

| NFR-ID | Category | Requirement / constraint | Target |
|---|---|---|---|
| NFR-04 | Reliability | Order, Customer and stock records shall remain available and recoverable. | system maintains at least 99% data availability through Supabase services. |
| NFR-05 | Compatibility | The application shall support commonly used Android devices. | Compatible with Android 8.0 (API 26) and above. |
| NFR-06 | Security & Privacy | Gemini API credentials shall be protected and customer data shall not be exposed. | API keys stored outside source code, no secrets committed to GitHub repository. |

**Table 7: Non-functional requirements and constraints**

### 5.5 System Environment

| Layer | Technology / tool | Justification |
|---|---|---|
| Mobile app | Flutter (Dart 3), Material 3 | Course requirement, cross-platform mobile application development and modern UI support |
| State management | Provider | Lightweight, easy to learn and suitable for managing application state in a small-to-medium sized project |
| Backend / database | Supabase (Authentication, Database, Storage) | Secure authentication, cloud database services, file storage and real time data synchronization for orders, stock records, customer details and feedback management |
| Maps Service | Google Maps API | Supports delivery location tracking and location visualisation for customers |
| AI service | Google Gemini API | Generates business insights based on sales and stock data to support inventory and sales management decisions |

| Layer | Technology / tool | Justification |
|---|---|---|
| Version control & CI | GitHub, GitHub Actions | Support collaborative development, version control, and automated build verification |
| Target devices | Android 8.0+ phones | Compatible with commonly used Android devices among the target users. |

**Table 8: System environment**

## 6. Task Allocation & Team Contribution

| Member | TSPi role | Owned features (FR-IDs) | Other responsibilities / contribution |
|---|---|---|---|
| DAMIA QISTINA BINTI FAIZAL | Software Project Manager | FR-01<br>FR-02<br>FR-03<br>FR-04 | Coordinates milestone planning, GitHub repository controls, client communication, weekly WEEK status reports, and risk escalation |
| SNG NIC CO | Development Manager/ Backend / AI Integration Specialist | FR-17<br>FR-18<br>FR-19<br>FR-20<br>FR-AI-01 | API integration, Google Gemini Generative AI connection, error/fallback handling, and LOGAI evidence |
| SITI AISYAH BINTI SABEKI | Lead Systems Analyst/ Requirements Engineer | FR-09<br>FR-10<br>FR-11<br>FR-12 | Leads requirements elicitation with Serooja (client), authors SRS, manages requirement IDs and traceability, and defines acceptance criteria |
| AINUR HANUM ABDULLAH TASIP | Quality Assurance & CI/CD Engineer | FR-13<br>FR-14<br>FR-15<br>FR-16 | Leads test planning/automation, GitHub Actions CI/CD workflow execution, defect tracking (LOGD), test execution logging |

| Member | TSPi role | Owned features (FR-IDs) | Other responsibilities / contribution |
|---|---|---|---|
| | | | (LOGTEST), and coverage reports. |
| SABRINA NATASHA BINTI SHUDARSAN | Lead Mobile Architect | FR-05<br>FR-06<br>FR-07<br>FR-08 | Leads UI/UX prototyping, Flutter state-management architecture, navigation structure, Dart modularity, and SDD drafting. |

**Table 9: Task allocation and contribution**

## 7. Team Goals

| No. | Team goal | Measure |
|---|---|---|
| G1 | Develop a quality mobile application that meets the client's requirements. | All Must-have FRs pass acceptance tests, and all identified defects are recorded and tracked. |
| G2 | Completed project tasks according to the planned schedule. | Review project progress during weekly meetings and complete all major milestones by their deadlines. |
| G3 | Ensure effective teamwork through clear role allocation and fair contribution among all team members. | Each member completes ≥1 assigned task per sprint and updates their task status on the project board. |
| G4 | Continuously improve the team's development process and product quality. | Conduct a postmortem at each milestone, review defects and progress data, and document at least one improvement action for the next cycle. |

**Table 10: Team goals**

## 8. Expected Results

### 8.1 Working System

From the client's side as the seller, KerepekHub is primarily used to manage and record orders received, verify payments made through external payment gateways, update order status, and record sales to monitor business performance. The customers on the other hand get to see product availability through stock management by the client, place their orders, complete payments through an external payment gateway, track order status, and leave feedback. The

application will also include an AI feature that analyzes sales and stock data for the seller dashboard to display sales trends and notify about product restocks. KerepekHub will be considered ready for release once all of the key features are implemented and the overall project successfully meets the agreed business requirements with a client UAT sign-off.

### 8.2 Documentation and Artefacts

| Deliverable | Submitted with |
|---|---|
| Project Proposal (original) | Proposal Submission (Week 4) |
| Revised Proposal and Software Requirements Specification (SRS) | Initial Report (Week 9) |
| Runnable prototype (4–5 screens), tagged release | Demo 1 (Week 9) |
| Software Design Specification (SDS), test plan and results (LOGTEST), user manual, maintenance plan, client acceptance form | Final Report (Week 13) |
| Release APK, demo video (5–10 min), presentation slides | Final Demo 2 (Week 14) |
| GitHub repository and complete project file | e-Archive (Week 15) |
| TSPi forms (LOGT, LOGD, LOGAI, SCHEDULE, SUMP, SUMQ, SUMS, SUMTASK, WEEK, PEER) and meeting minutes | Ongoing; compiled in Final Report |

**Table 11: Deliverables**

## 9. Project Planning

### 9.1 Methodology

KerepekHub is built using Scrum with 2-weeks sprints. Scrum works well for the customer since the owner is a small firm, and once the owner sees the app functioning, requirements like product descriptions, order statuses, and dashboard requirements are likely to change. At the end of every sprint, the team demonstrates the current version of the application to the business owner and collects feedback to guide future enhancements. The team follows TSPi roles, with the Project Manager coordinating team activities, the development manager handles backend development and AI integration, the Lead Systems gathers and analyses client requirements, the Quality Assurance ensures software quality through testing and automated workflows and the Lead Mobile Architect oversees the mobile application's architecture and design. Each feature is recorded as a GitHub issue and implemented in a dedicated branch. A teammate then reviews the changes through a pull request. Automated tests are run before the code is merged, ensuring that only code that passes the tests is integrated into the main branch.

### 9.2 Submission Schedule

| Phase | Deliverable | Week | Due |
|---|---|---|---|
| Phase 1: Planning & Development | Proposal Submission | 4 | 16 Oct 2026 |
| | Proposal Presentation | 5 | 19 – 23 Oct 2026 |
| | Initial Report | 9 | 16 Nov 2026 |
| | Demo 1 | 9 | 16 – 20 Nov 2026 |
| Phase 2: Finalisation & Closure | Final Report | 13 | 24 Dec 2026 |
| | Final Demo 2 | 14 | 28 – 31 Dec 2026 |
| | e-Archive | 15 | Before 10 Jan 2027 |

**Table 12: Submission schedule**

### 9.3 Development Milestones (GitHub)

| Milestone | Week | Key output | Status |
|---|---|---|---|
| M1 – Scope & Requirements | 4 – 5 | GitHub repo, issue board, CI file; proposal and initial SRS in /docs | Completed |
| M2 – Design & Prototype | 9 | SDS draft, 4–5 screen prototype, green CI, v0.2.0 tag | Pending |
| M3 – Product & Verification | 13 | ≥50 tests, ≥60% coverage, AI feature working | Pending |
| M4 – Final Release & Acceptance | 15 | Release APK v1.0.0, client sign-off, e-Archive | Pending |

**Table 13: Development milestones**

*▸ Insert a Gantt chart (Figure 3) from Week 3 to Week 15 showing sprints, the seven submission dates and the four milestones. It should match your TSPi TASK and SCHEDULE worksheets.*

Figure 3: Gantt chart

### 9.4 Risks and Assumptions

| ID | Risk / assumption | Likelihood | Impact | Mitigation |
|---|---|---|---|---|
| R1 | Client unavailable for meetings | Medium | High | Agree fixed fortnightly slot at the sprint review by using a Whatsapp voice or video call as a fallback |
| R2 | AI API cost limits so the sales and stock analysis stops working | Medium | High | Send only summarised sales and stock data rather than the raw records |
| R3 | The team is still learning Flutter which slows development | Medium | High | Assign work by strength and keep notes on solutions in Github |

**Table 14: Risks and assumptions**

## 10. References

▸ *APA 7th edition. Include client documents, libraries, packages and APIs used. Give enough detail for a reader to find each source.*

## Appendix A: Client Meeting Evidence

- *Meeting minutes summary (date, attendees, key decisions) and, if allowed, a photo with the client.*




## Appendix B: Revision History

| Version | Date | Changes (e.g. after proposal presentation feedback) | By |
|---|---|---|---|
| *1.0* | 16/10/26 | *Original submission* | |
| *1.1* | | | |